Hoapa for Logicians

This document is a typeset transcription of the handwritten note `hoapa_for_logicians.jpg`. The spatial layout has been normalized for readability. Question marks and the final editorial notes record marks whose reading or intended meaning remains uncertain.

Universe of contexts

$$\mathsf{UC} = \text{universe of contexts}, \qquad \mathsf{UC}_n = \{\mathsf{Ctx}, \mathsf{Ag}, \mathsf{Sym}\}.$$

Here

$$\begin{aligned} \mathsf{Ctx} &= \text{context } n, \\ \mathsf{Ag} &= \text{agents (unityped)}, \\ \mathsf{Sym} &= \text{set of symbols}. \end{aligned}$$

The upper-right sketch gives the emergence sequence

$$\mathsf{Ctx} \xrightarrow{\text{emergence}} \mathsf{Ag} \xrightarrow{\text{emergence}} \mathsf{Sym}.$$

The set of symbols is used for *typed documentation*. Such documentation is about the context and is produced by the agents; it gives rise, by another emergence step, to the next universe of contexts:

$$\text{typed documentation} \xrightarrow{\text{emergence}} \mathsf{UC}_{n+1}.$$

Example in nature

$$\begin{aligned} \mathsf{Ctx} &= \text{chemistry on Earth}, \\ d\text{-}c &= \text{phenotypes}, \\ \mathsf{Ag} &= ?, \\ \mathsf{Sym} &= \text{ATCG}. \end{aligned}$$

Gills document / prove / witness water.

"Theory of everywhere"

The central evolution chain is

$$\mathsf{UC}_n \xrightarrow{\mathsf{Ag}} \mathsf{Doc}^{\mathrm{TT}} \xrightarrow{\mathsf{Evol}} \mathsf{UC}_{n+1}.$$

It is annotated as *not a series, but a network*.

The page then groups the following terms as a category, with the annotation *objects, morphisms*:

$$\underbrace{\{\mathsf{Evol}, \mathsf{UC}_n, \mathsf{Hoapa}\}}_{\text{category}} \qquad \text{(objects, morphisms)}.$$

Condition

Ctx endowed with perspectivalism / modalities.

Editorial notes

- The handwritten *d-c* in the natural example is legible as those characters, but its expansion is not given.
- $\mathsf{Doc}^{\mathrm{TT}}$ is the most likely reading of the middle object in the evolution chain; the superscript is handwritten as TT.
- The braces around $\{\mathsf{Evol}, \mathsf{UC}_n, \mathsf{Hoapa}\}$ and the label *category* preserve the page grouping without inferring which listed terms are intended as objects and which as morphisms.